

BANARAS HINDU UNIVERSITY

Under Graduate Semester III Examination, 2025-26 (NEP)

COMPUTER SCIENCE**Paper Code: CSCMD31 : Introduction of Cyber Security**

Time: 2:30 Hours

Full Marks: 70

REF NO. CECNF/N/099

*Instructions: Attempt any **five** questions. Question No. 1 is **compulsory**. All questions carry equal marks (14 marks each). Write your Roll No. at the top immediately on receipt of this question paper.*

1. Objective Type Questions: $[7 \times 2 = 14]$

- (a) What is the most suitable definition of Cyber Security?
 - (i) The practice of only protecting computer hardware from physical damage.
 - (ii) Using cryptography to secure online banking transactions.
 - (iii) Protecting digital systems, networks, programs, and data from attacks, damage, or unauthorized access.
 - (iv) The process of developing anti-virus software.
- (b) Which term in the CIA Triad ensures that information is accurate and unaltered?
 - (i) Confidentiality
 - (ii) Integrity
 - (iii) Availability
 - (iv) Authentication
- (c) Which type of malware is a standalone program that is self-replicating and spreads automatically across networks by exploiting system vulnerabilities without requiring human intervention?
 - (i) Virus
 - (ii) Trojan
 - (iii) Worm
 - (iv) Spyware
- (d) What is the primary function of Ransomware?
 - (i) To secretly record every keystroke made on the infected system.
 - (ii) To disguise itself as legitimate software to create a backdoor for attackers.

- (iii) To encrypt a victim's files, making them inaccessible until a ransom is paid.
 - (iv) To continuously monitor user activity and relay personal information to an attacker.
- (e) What is the term for a scam attack that uses fraudulent voice calls to manipulate a victim into giving up sensitive information?
- (i) Phishing
 - (ii) Smishing
 - (iii) Baiting
 - (iv) Vishing
- (f) Which type of attack involves a single source flooding a target server with traffic until it is overwhelmed and cannot serve legitimate users?
- (i) Malware
 - (ii) MitM (Man-in-the-Middle)
 - (iii) DoS (Denial-of-Service)
 - (iv) Business Email Compromise (BEC)
- (g) Which of the following is defined as a weakness in a system or process that a threat could potentially exploit?
- (i) Asset
 - (ii) Risk
 - (iii) Task
 - (iv) Vulnerability
2. Define Cyberspace and Cyber Crime? List and explain the key criteria to choose a web browser. [14]
3. Discuss the guidelines for setting a secure password. What is Two-step authentication? How can a Password Manager be useful to manage passwords? [14]
4. Write the best-practice guidelines for safer social media. Do you agree that Smartphone Security is very crucial in Cyber-security? How will you protect your smartphone devices from attack landscapes? [14]
5. What is IT Act 2000? List the various sections of this act related to Cyber Security and discuss any one section with its sub-section in brief. [14]
6. Discuss Cyber Security Assurance. What do you understand by Vulnerability Assessment and Penetration Testing? What are the key challenges in Cyber Security Assurance to achieve? Explain in brief. [14]

7. Compare any **two** of the following terminologies:

[2 × 7 = 14]

- (a) Confidentiality and Integrity
- (b) Online Banking and Mobile Banking
- (c) Ethics and Policies of Computer
- (d) Section 70 and Section 66 of IT Act